RAJALAKSHMI ENGINEERING COLLEGE

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D2

Started on	Sunday, 30 August 2026, 2:34 PM
State	Finished
Completed on	Monday, 14 September 2026, 9:07 PM
Time taken	15 days 6 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int count(int arr[], int left, int right, int num)
{
    int c = 0;

    for (int i = left; i <= right; i++)
    {
        if (arr[i] == num)
            c++;
    }

    return c;
}

int majorityElement(int arr[], int left, int right)
{
    // Only one element
    if (left == right)
        return arr[left];

    int mid = (left + right) / 2;

    // Find majority element in left and right halves
    int leftMajority = majorityElement(arr, left, mid);
    int rightMajority = majorityElement(arr, mid + 1, right);

    // If both halves give the same element
    if (leftMajority == rightMajority)
        return leftMajority;

    // Count both candidates in the complete range
    int leftCount = count(arr, left, right, leftMajority);
    int rightCount = count(arr, left, right, rightMajority);
```

```
35
36      if (leftCount > rightCount)
37          return leftMajority;
38      else
39          return rightMajority;
40  }
41
42  int main()
43  {
44      int n;
45      scanf("%d", &n);
46
47      int arr[n];
48
49      for (int i = 0; i < n; i++)
50          scanf("%d", &arr[i]);
51
52      printf("%d", majorityElement(arr, 0, n - 1));
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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